

Task 1 Interim Report

Brent Oil Price Analysis: Change Point Detection Foundation
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1. Planned Analysis Steps

- Phase 1: Data Preparation and Exploration
- Load Brent oil price data (1987-2022) and perform quality checks
 - Conduct time series analysis: trend, stationarity, volatility patterns
 - Generate initial visualizations and statistical summaries
- Phase 2: Event Data Integration
- Compile major geopolitical events, OPEC decisions, and economic shocks
 - Create structured dataset with dates, descriptions, and categorization
 - Align event data with price timeline for correlation analysis
- Phase 3: Bayesian Change Point Modeling
- Design Bayesian change point model using PyMC framework
 - Define prior distributions and likelihood functions
 - Implement MCMC sampling with convergence diagnostics
- Phase 4: Analysis and Interpretation
- Detect structural breaks and map to historical events
 - Quantify impact magnitude and duration of price changes
 - Conduct validation and sensitivity analysis
- Phase 5: Communication and Visualization
- Create interactive dashboard for stakeholder insights
 - Generate comprehensive reports with actionable recommendations

2. Structured Event Dataset

- Dataset Overview:**
- Total Events: 31 major oil market events (1990-2022)
 - Data Fields: Date, Event, Event_Type, Severity, Description, Duration
 - Event Categories: Geopolitical Conflict, OPEC Decision, Economic Shock, Natural Disaster, etc.
 - Severity Levels: High (19), Medium (10), Low (2)

Sample Events Dataset:

Date	Event	Type	Severity
1990-08-02	Iraq Invasion of Kuwait	Geopolitical Co...	High
1991-01-17	Gulf War Operation Desert Stor...	Geopolitical Co...	High
1997-07-01	Asian Financial Crisis	Economic Shock	Medium
1998-12-30	OPEC Production Cut Agreement	OPEC Decision	Medium
2001-09-11	9/11 Terrorist Attacks	Geopolitical Co...	High
2003-03-20	Iraq War Invasion	Geopolitical Co...	High
2005-08-29	Hurricane Katrina	Natural Disaste...	High
2008-07-03	Oil Price Peak (\$147/barrel)	Market Event	High
2008-09-15	Global Financial Crisis Lehman...	Economic Shock	High
2010-04-20	Deepwater Horizon Oil Spill	Environmental D...	Medium
2011-02-15	Arab Spring Libya Conflict	Geopolitical Co...	High

2011-03-11	Fukushima Nuclear Disaster	Natural Disaste...	Medium
2012-01-01	EU Sanctions on Iran	International S...	Medium
2014-06-01	OPEC Production Increase	OPEC Decision	High
2014-11-27	OPEC Refuses Production Cut	OPEC Decision	High

Event Type Distribution:

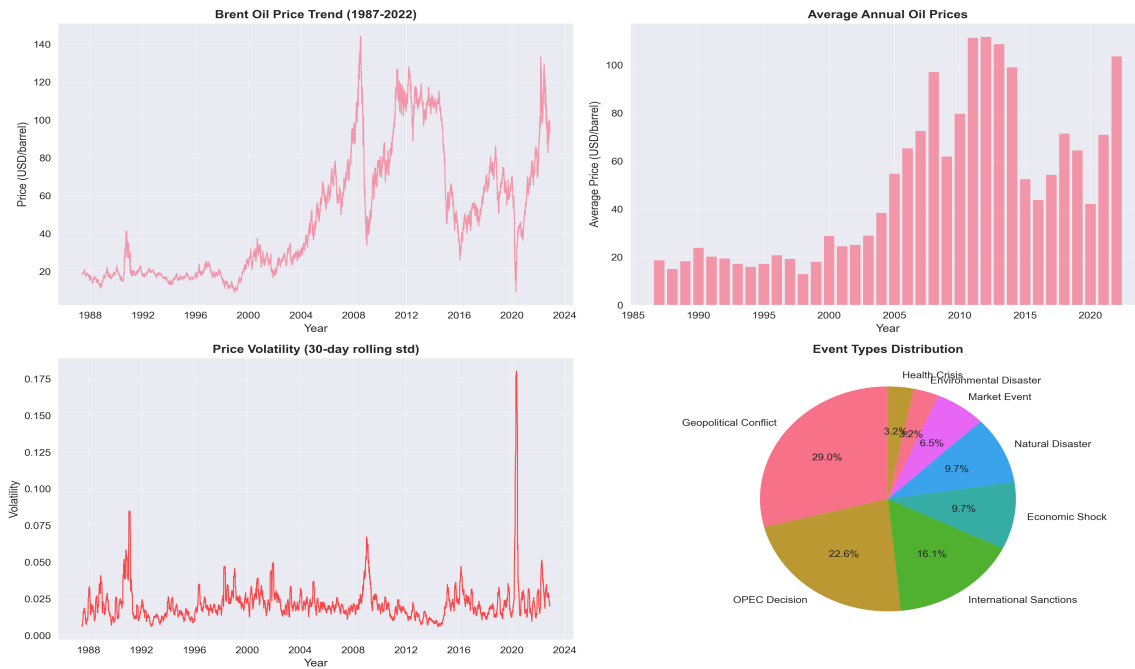
Event Type	Count
Geopolitical Conflict	9
OPEC Decision	7
International Sanctions	5
Economic Shock	3
Natural Disaster	3

3. Initial EDA Findings

Dataset Overview:

- Total Records: 9,011 daily price observations
- Time Period: May 20, 1987 to November 14, 2022 (35+ years)
- Price Range: \$9.10 to \$143.95 per barrel
- Mean Price: \$48.42 (Median: \$38.57, Std Dev: \$32.86)
- Right-skewed distribution (mean > median)
- High volatility with average daily volatility of 2.1%
- Multiple extreme price events identified
- Clear non-stationarity requiring specialized modeling
- 1990: \$14.68-\$41.45 (Gulf War impact)
- 2008: \$33.73-\$143.95 (Financial crisis peak)
- 2020: \$9.12-\$70.25 (COVID-19 pandemic)
- 2022: \$78.25-\$133.18 (Russia-Ukraine conflict)
- High-severity events: 19 total
- Average price change: -5.73% around events
- Largest positive: +35.36% (Iraq Invasion 1990)
- Largest negative: -34.23% (OPEC+ Price War 2020)
- Highest volatility: May 6, 2020 (COVID period)
- Clear patterns of high volatility during crisis periods
- Sustained volatility during geopolitical conflicts
- Spillover effects across related market events

Key Visualization: Price Analysis Overview



Conclusion

Task 1 has successfully established the foundation for Bayesian change point analysis of Brent oil prices. The comprehensive dataset spans 35+ years with 31 major events categorized by type and severity. Initial EDA reveals significant price volatility (\$9.10-\$143.95 range) with clear correlations to geopolitical and economic events. The 5-phase analysis workflow provides a structured approach for implementing robust change point detection models. The foundation is complete for proceeding with Bayesian analysis implementation.